

Physics of Complex Systems Homework 9

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Problem 1 (Density Matrix of a Single Qubit). For a given ensemble of qubits the observables

$$A = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & 2i \\ -2i & 0 \end{pmatrix},$$

have been measured many times, giving the expectation values

$$\langle A \rangle = 2, \quad \langle B \rangle = \frac{1}{2}, \quad \langle C \rangle = 0.$$

- (a) Determine the density matrix of the ensemble.
- (b) Is the ensemble pure or mixed?
- (c) If we measure σ^z , what is the probability to get +1?
- (d) Determine the expectation values

$$\langle \sigma^x \rangle, \quad \langle \sigma^y \rangle, \quad \langle \sigma^z \rangle.$$

Proof. (a) Given a density matrix ρ , the expectation value of an operator $\mathbf{A}$ is given by $\langle \mathbf{A} \rangle = \text{Tr}(\rho \mathbf{A})$. We implement the trace 1 and hermiticity conditions by defining our density matrix using

$$\rho = \begin{pmatrix} a & b \\ b^* & 1-a \end{pmatrix}.$$

Substituting this in, we find

$$\langle \mathbf{A} \rangle = 1 + 2a = 2$$

$$\langle \mathbf{B} \rangle = -1 + 2a + b + b^* = \frac{1}{2}$$

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$$\langle \mathbf{C} \rangle = -2ib + 2ib^* = 0$$

From the third equation, we see that $b = b^*$, i.e. b is real. From the first equation, we can see easily that $a = \frac{1}{2}$. Substituting this in to the second equation, we find that $2b = \frac{1}{2}$, or $b = \frac{1}{4}$.

(b) We do this using the trace criterion: We have

$$\rho^2 = \begin{pmatrix} \frac{5}{16} & \frac{1}{4} \\ \frac{1}{4} & \frac{5}{16} \end{pmatrix}.$$

Thus, the trace is

$$\text{Tr}(\rho^2) = \frac{10}{16} \neq 1.$$

The ensemble is mixed.

(c) ..

(d) We compute

$$\rho\sigma^z = \begin{pmatrix} \frac{1}{2} & -\frac{1}{4} \\ \frac{1}{4} & -\frac{1}{2} \end{pmatrix}.$$

Thus, we have

$$\langle \rho\sigma^z \rangle = 0.$$

Similarly, we have

$$\rho\sigma^x = \begin{pmatrix} \frac{1}{4} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{4} \end{pmatrix}$$

with

$$\langle \rho\sigma^x \rangle = \frac{1}{2}$$

and

$$\rho\sigma^y = \begin{pmatrix} \frac{i}{4} & -\frac{i}{2} \\ \frac{i}{2} & -\frac{i}{4} \end{pmatrix}$$

with

$$\langle \rho\sigma^y \rangle = 0.$$

□

Problem 2 (Criterion for the Equivalence of Ensembles). Consider a statistical ensemble $\{|\psi_i\rangle, p_i\}$ with the associated density matrix

$$\rho = \sum_{i=1}^N p_i |\psi_i\rangle \langle \psi_i|.$$

Here N is arbitrary (smaller or larger than the dimension of the Hilbert space) and the vectors $|\psi_i\rangle$ are pairwise different and normalized but not necessarily orthogonal. Moreover, let

$$\rho = \sum_{k=1}^K \lambda_k |k\rangle \langle k|$$

be the spectral decomposition of ρ with orthonormal eigenvectors $|k\rangle$ and eigenvalues λ_k .

The aim of this exercise is to work out an equivalence theorem for different ensembles with the same density matrix.

- (a) Show that every vector $|\psi_i\rangle$ can be written as a linear combination of the eigenvectors $|k\rangle$, i.e.

$$|\psi_i\rangle \in \text{span}\{|k\rangle\}.$$

- (b) Use (a) to prove that every $|\psi_i\rangle$ can be written as

$$\sqrt{p_i} |\psi_i\rangle = \sum_{k=1}^K c_{ik} \sqrt{\lambda_k} |k\rangle,$$

with coefficients c_{ik} obeying

$$\sum_{i=1}^N c_{ik} c_{il}^* = \delta_{kl}, \quad k, l = 1, \dots, K.$$

- (c) Because of (a) we know that $N \geq K$. For $N = K$ the result of (b) would imply that the coefficient matrix c_{ik} is unitary. Show that for $N > K$ the rectangular matrix c_{ik} can be extended to a square unitary matrix such that the relation in (b) still holds.
- (d) So far we have proven that if

$$\sum_{i=1}^N p_i |\psi_i\rangle \langle \psi_i| = \sum_{k=1}^K \lambda_k |k\rangle \langle k|,$$

then there exists a unitary $N \times N$ matrix c_{ik} such that

$$\sqrt{p_i} |\psi_i\rangle = \sum_{k=1}^N c_{ik} \sqrt{\lambda_k} |k\rangle.$$

Prove the converse statement.

(e) Use (a)–(d) to show that

$$p_i = \sum_{k=1}^N T_{ik} \lambda_k,$$

(or in vector form $\vec{p} = T\vec{\lambda}$), where T is a doubly-stochastic matrix, i.e. all matrix entries are real and nonnegative and

$$\sum_{i=1}^N T_{ik} = \sum_{k=1}^N T_{ik} = 1.$$

(f) Use e.g. Mathematica to work out the explicit example

$$p_1 = p_2 = \frac{1}{2}, \quad |\psi_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |\psi_2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

with $N = K = 2$, and determine $\lambda_{1,2}$, $|\phi_{1,2}\rangle$, c_{ij} , and T_{ij} . Check the properties derived in (e).

Proof. (a) This follows from the spectral theorem; since ρ is Hermitian, the eigenvectors $|k\rangle$ form an orthogonal decomposition of the Hilbert space.

□